

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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SIMATS Online Programming Lab — 100 Perfect Squares — 100 TOPICS

100 TOPICS

Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10.

Sample Input & Output:

Enter lower range: 1

Enter upper range: 40

[1, 4, 9, 16, 25, 36]

Test case:

1. Enter lower range: 50

Enter upper range: 100

2. Enter lower range: 5

Enter upper range: 8

3. Enter lower range: 10

Enter upper range: 5

4. Enter lower range: 500

Enter upper range: 500

5. Enter lower range: 0

Enter upper range: -100

Testcase:

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int low=sc.nextInt();
6         int up=sc.nextInt();
7         List<Integer> res=new ArrayList<>();
8         for(int num=low;num<=up;num++){
9             int sq=(int)Math.sqrt(num);
10             boolean per=(sq*sq==num);
11             if (per){
12                 int sum=sum(num);
13                 if (sum<10){
14                     res.add(num);
15                 }
16             }
17         }
18     }
19 }
```

Output

```
[1, 4, 9, 16, 25, 36]
```

Q1
NumbersQ2
Control

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Develop a JAVA code to display the balance. Include the following members:

- Design a class to represent a bank account.
- Data Members: Name of the depositor, Account number, Type of account(Savings/Current), Balance amount in the account(Minimum balance is Rs.500.00)
- Methods:

1. To read account number, Depositor name, Type of account,
2. To deposit an amount (Deposited amount should be added with it)
3. To withdraw an amount after checking balance(Minimum balance must be Rs.500.00)

Note : Assume that balance amount = 10000

Testcase: 1. 100, Raja, S, 8000,

2. Raja, 100, S, 9000

3. 101, Rani, S, 12000

4. 102, Raghu, W, 8000

5. 103, Ravi, C, 10000

Your Code

```
1 import java.util.*;
2 class Bankacc{
3     String name;
4     int accno;
5     char type;
6     double bal;
7     void read(int accno, String name, char type, double bal) {
8         this.accno = accno;
9         this.name = name;
10        this.type = type;
11        this.bal = bal;
12    }
13    void deposit(double amt) {
14        bal+=amt;
15    }
16    void withdraw(double amt) {
17        if (bal-amt < 500) {
18            System.out.println("Min balance must be atleast 500");
19        } else {
20            bal-=amt;
21        }
22    }
23 }
```

Output

```
Account no: 100
Name: kumar
Type: S
Balance: 10000.0
Account no: 100
```